

Python Security Log Parser & Alert Generator

Simple rule-based automation for authentication-event triage

Purpose

Parse a synthetic CSV of authentication events and generate JSON alerts for repeated failures and failures followed by success. This demonstrates automation fundamentals that complement SIEM work; it is not presented as a replacement for production detection platforms.

Alert type	Severity	Detection logic	Analyst value
Repeated Authentication Failures	Medium	5 failures for same user + source within 5 minutes	Surfaces potential guessing or misconfiguration for triage.
Failures Followed by Success	High	>=3 recent failures followed by success from same user + source	Raises priority because account compromise becomes plausible.

Generated Output Summary

The sample run generated 2 JSON alert records from the included authentication dataset.

Type	Severity	User	Source	Evidence
Repeated Authentication Failures	medium	j.smith	203.0.113.44	5 failures within 5 minutes
Failures Followed by Success	high	j.smith	203.0.113.44	8 recent failures followed by success

Files

- python/security_log_parser.py
- data/auth_events_for_parser.csv
- outputs/generated_alerts.json